

535/1

Physics

Paper 1

$2\frac{1}{2}$ hours



Apex Examination Council

MOCK EXAMINATIONS 2025

Uganda certificate of education

PHYSICS

Paper 1

2 hours 30 Minutes

INSTRUCTIONS TO CANDIDATES;

This paper consists of **two** sections; **A** and **B**. It has **seven** examinations items.

Section **A** has **three compulsory** items.

Section **B** has **two** parts; **I** and **II**. Answer **one** item from **each** part.

Answer **five** items in all.

Answers to all items **must** be written on the answer sheet(s) provided.

SECTION A

(Answer all items in this section)

ITEM 1

Tourism is one of the most improved sectors and it was to my view to tour during our honey moon with my wife. It was a very cool morning when we decided to board a car that took us a very long journey to reach the site. On our way while driving, we could see a pool of water in front of us on the tarmac road which could disappear as we approach it. This surprised us so much. On reaching the shores of Lake Victoria, the tour guide told us not to step in some parts of the water because some are deep but they appear shallower which we denied and called it a lie because how could someone see and say you have seen wrongly?

As I was moving on the shores, the points of waters are widely spaced and some points of wader waves were near each other, when I asked the guide, he told me it is due to difference in whether condition. We continued enjoying our day till night time when my wife told me that she is going for a short call just to release after an hour that she is not back yet. I tried to call her but my voice was repeating it's self that I realised which was 330m from a tall hill.

The tour guide got a torch which was giving out many colours, red yellow and green which could change the colour of my shirt with blue strips and that surprised me so much. Praise God my wife came back but she had picked a certain mirror which made out heads so big and we were looking so ugly which made us to laugh and we thought we had a magic mirror.

HINT: Speed of sound in air is 330 ms^{-1}

Task

As a physics student, explain why;

- i) The meaning of the pool of water
- ii) Why the tour guide refused the couple not to step at each part of water
- iii) Explain the repeating of sound as heard by the man and what would the sound take to come back to him.

- iv) Was the difference in weather causing changes in water waves?
- v) Which colours did the man's shirt change to and justify about the mirror picked by the woman

Item 2

Each year the rate at which accidents are made is on a rise and many people came up with serious injuries. One day, my neighbour got an accident and he got serious injuries which made us to rush him to the hospital. They recommended us to take him to a machine which examined him and we were given the results on a photographic film.

Many of us did not know how to interpret the results. I was also requested to enter and they examine me which the doctor refused but I did not know why?

There came a lady who was putting on a neckless designed in bones of ancient animals and we were surprised and we could not estimate the time those bones had spent on earth. We were sent back from the hospital but everyone went surprised about many things.

Task

As a physics student, explain to them;

- i) How the machine operates
- ii) Why the doctor refused him to enter for an examination too.
- iii) How to interpret the results on photographic film.
- iv) How could they determine the age of the bones above.

Item 3

One of the misguided information by our ancestors comes from space where many people don't know what place up there. When we were still young even up to now people don't know what brings a burning stone (shooting star) but rather just ask for blessing.

One day, I asked my group members how the earth came into existence which many of us could not even try to imagine. After failing that, we asked ourselves how the sun could produce energy and why the earth is the best habitat for man. Who knows why the stars twinkle? Why do they produce different colours, and why some are brighter than the others? Many questions are asked but there are few questions:

Task

As a physics student in your school, explain all the above misunderstandings in order to give the lay man a clear satisfaction.

SECTION B

PART I;(Answer one item from this part)

Item 4.

In Uganda, many people go for sauna and steam bath. One day I visited there but they could pass steam at 100°C into a bath tub of 150g containing 340g of water at 15°C . This is done until the temperature of the steam bath and its contents is found to be 52.5°C . I could not tell the latent heat of vaporization of water. From there, I entered a car to drive back home but I used a certain speed because I was out of time. Unfortunately, the traffic officer stopped my car but due to a sudden stop, my body moved towards the steering wheel which made my tyres to wear. The traffic officer had a machine which could measure the temperature of the tyre to prove him whether it was driving at disallowed speed and if the temperature exceeds 40°C you are bounded a fine. The instruments gave out results;

HINT:

Specific heat capacity of copper $400\text{Jkg}^{-1}\text{K}^{-1}$

The length at ice point was 22cm

The length at steam point was 62cm.

The length at unknown temperature = 42cm.

Task

As a physics student;

- i) What could be the latent heat of vaporization?
- ii) Explain why his body moved towards the steering
- iii) What do you think brought the wear of the tyre?
- iv) Was he given a fine?

Item 5

It was a rainy day when a couple was moving in a muddy area heading to a tall mountain, following closely the woman's high hill shoe would penetrate the mud but the man's could not yet the woman was small and the man was fat.

On climbing the mountain, they had an instrument which they failed to understand but it could read a pressure of 75cm/Hg at the bottom of the mountain and 73.5cm/Hg at the top of the mountain. On reaching the top of the mountain, they started preparing something to eat but they were surprised why cooking takes a lot of time on top of the mountain and because of that they advised them to use a pressure cooker.

HINT

Density of mercury = 13600kgm^{-3}

Acceleration due to gravity = 10ms^{-2}

The density of air = 1.25kgm^{-3}

Task

Explain;

- i) Why the woman's shoe acted differently from the man's shoe
- ii) Which instrument do you think they used and how would someone determine the height they climbed.
- iii) Explain why cooking takes long on top of a mountain
- iv) How does a pressure cook work.

PART II; (answer one item from this part)

Item 6

One of the most disappointing thing on earth to the youth is when they are watching football and the power goes off. In our home village, we have a solution to that because we use a certain machine to produce electricity which gives out a loud sound though it is 80% efficient and has an input power of 3000W. In that football hall, we have two bulbs of 40W each and work for 8 hours a day.

HINT

Cost of one unit is 950/=

Task

- i) How the machine does operates?
- ii) Find out the power output of the machine
- iii) What do you think is the cost of electricity for the bulbs used for a month?
- iv) How can they minimise wastage of electricity?

Item 7

- a) Our neighbours have speakers in their rooms which make noise for us the whole night and due to that we are to vacate the room for another place. One time when I went to their room I saw that the instrument was labelled 1000W, 240V.

Task

- i) Explain how the instrument which makes noise works
 - ii) How can one know the current used by the instrument?
- b) The supplied current to the instrument is alternating current by the speaker uses direct current and we wonder how they makes it to change the current supplied to the required one by the speaker.

Task:

As a physic student, explain how it is possible

END